Engineering Mathematics III

ENGE702

Lecture 12: Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

Before we start on eigenvalues and eigenvectors, we will review some basic matrix arithmetic.
A *matrix* is a rectangular array of (real) numbers.

For example, the following are all matrices:

$$\begin{pmatrix} 2 & -1 & 0 \\ 3 & \sqrt{2} & \frac{1}{2} \end{pmatrix} \quad \begin{pmatrix} 9 & -2 & 0 \\ 2 & 1 & -3 \\ 5 & -1 & 2 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 6 \\ -1 \end{pmatrix}$$

We can add (and subtract) matrices, and multiply matrices by a scalar.

Two matrices can be added if they are the same size (same order). If they are not the same size, we cannot add them.

If two matrices are the same size, we add them by adding the corresponding coordinates.

Eigenvalues and Eigenvectors

Examples:

$$\begin{pmatrix} 2 & 1 & -3 \\ 0 & -2 & 5 \end{pmatrix} + \begin{pmatrix} -1 & 3 & 2 \\ 1 & 0 & -3 \end{pmatrix} =$$

$$\begin{pmatrix} 2+(-1) & 1+3 & (-3)+2 \\ 0+1 & (-2)+0 & 5+(-3) \end{pmatrix} = \begin{pmatrix} 1 & 4 & -1 \\ 1 & -2 & 2 \end{pmatrix}.$$

$$\begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 2 \\ 5 & 0 \end{pmatrix} = \begin{pmatrix} 2-4 & 0-2 \\ -1-5 & 3-0 \end{pmatrix} = \begin{pmatrix} -2 & -2 \\ -6 & 3 \end{pmatrix}$$

We can always multiply a matrix by a scalar. Just like with vectors, we simply multiply each component by the scalar.

Examples:

$$5\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} (5)(2) & (5)(1) \\ (5)(0) & (5)(3) \end{pmatrix} = \begin{pmatrix} 10 & 5 \\ 0 & 15 \end{pmatrix}$$

Eigenvalues and Eigenvectors

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$$\frac{1}{3}\begin{pmatrix} 1 & 7 \\ 6 & -2 \end{pmatrix} = \begin{pmatrix} (\frac{1}{3})(1) & (\frac{1}{3})(7) \\ (\frac{1}{3})(6) & (\frac{1}{3})(-2) \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{7}{3} \\ 2 & -\frac{2}{3} \end{pmatrix}$$

Eigenvalues and Eigenvectors

Matrix multiplication is a bit more complicated that the first two operations. At first it may seem a bit strange.

If we have two matrices A and B, where A is a $p \times q$ matrix and B is an $r \times s$ matrix, we can form the product AB only if $q = r$, ie only if the number of columns in A is the same as the number of rows in B.

For example, we can find the product $\begin{pmatrix} 1 & 0 & 4 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -3 & 6 \\ 2 & 1 \end{pmatrix}$,

but we cannot find the product $\begin{pmatrix} 2 & 0 & 1 \\ 2 & -6 & 3 \end{pmatrix} \begin{pmatrix} -4 & 2 & 1 \\ 5 & -2 & 7 \end{pmatrix}$.

If A is a $p \times q$ matrix and B is a $q \times s$ matrix, then AB will be a $p \times s$ matrix.

Eigenvalues and Eigenvectors

Suppose A is a $p \times q$ matrix and B is a $q \times s$ matrix.

To calculate the product AB, it can be helpful to view A as a collection of row vectors, and B as a collection of column vectors, so

$$A = \begin{pmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_p \end{pmatrix}, \text{ where } \mathbf{r}_i = \begin{pmatrix} a_{i1} & a_{i2} & \dots & a_{iq} \end{pmatrix}, \text{ and}$$

$$B = \begin{pmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_s \end{pmatrix}, \text{ where } \mathbf{c}_i = \begin{pmatrix} b_{1i} \\ b_{2i} \\ \dots \\ b_{qi} \end{pmatrix}$$

Then to find the entry c_{ij} in AB, we take the dot product $\mathbf{r}_i \cdot \mathbf{c}_j$

This gives us every entry in the $p \times s$ matrix AB.

Eigenvalues and Eigenvectors

Example:

Find $\begin{pmatrix} 0 & 2 & -1 \\ 3 & 4 & -2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 0 \\ -2 & 3 \end{pmatrix}$

Eigenvalues and Eigenvectors

Example:

Find $\begin{pmatrix} 0 & 2 & -1 \\ 3 & 4 & -2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 0 \\ -2 & 3 \end{pmatrix}$

First, check that the number of columns in the matrix on the left is equal to the number of rows in the matrix on the right. There are 3 of each, so that is OK, we can perform the calculation.

Now work out the size of the product. The matrix on the left has 2 rows, and the matrix on the right has 2 columns, so the product is 2×2.

To find c_{11}, the top left entry in the product, we take the dot product of the first (top) row in the left matrix with the first (left) column in the right matrix. This gives us

$$\begin{pmatrix} 0 & 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} = (0)(2) + (2)(3) + (-1)(-2) = 0 + 6 + 2 = 8$$

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Eigenvalues and Eigenvectors

Now find c_{12}. This is the first (top) row on the left with the second (right) column on the right. We have:

$$\begin{pmatrix} 0 & 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = (0)(-1) + (2)(0) + (-1)(3) = 0 + 0 - 3 = -3$$

Now find c_{21}. This is the second (bottom) row on the left with the first (left) column on the right. We have:

$$\begin{pmatrix} 3 & 4 & -2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} = (3)(2) + (4)(3) + (-2)(-2) = 6 + 12 + 4 = 22$$

Now find c_{22}. This is the second (bottom) row on the left with the second (right) column on the right. We have:

$$\begin{pmatrix} 3 & 4 & -2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = (3)(-1) + (4)(0) + (-2)(3) = -3 + 0 - 6 = -9$$

Eigenvalues and Eigenvectors

Now we have all components in the product. Putting it all together:

$$\begin{pmatrix} 0 & 2 & -1 \\ 3 & 4 & -2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 0 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 8 & -3 \\ 22 & -9 \end{pmatrix}$$

Eigenvalues and Eigenvectors

Example:

Find $\begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

Eigenvalues and Eigenvectors

Example:

Find $\begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

The number of columns on the left equals the number of rows on the right, so we can perform the calculation. The number of rows on the left is 2 and there is 1 column on the right, so the product will be a 2×1 matrix.

To find c_{11}, we find

$$\begin{pmatrix} 1 & 4 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \end{pmatrix} = (1)(-2) + (4)(1) = -2 + 4 = 2$$

To find c_{21}, we find

$$\begin{pmatrix} 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \end{pmatrix} = (2)(-2) + (-1)(1) = -4 - 1 = -5$$

So $\begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$.

Eigenvalues and Eigenvectors

If A is a matrix, then the **transpose** of A, denoted A^T, is obtained by interchanging the rows of A with the columns of A. So the first (top) row becomes the first (left) column, the second row becomes the second column etc. For example:

If $A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 2 & 1 \end{pmatrix}$ then $A^T = \begin{pmatrix} 2 & 0 \\ 1 & 2 \\ 3 & 1 \end{pmatrix}$.

If $B = \begin{pmatrix} 0 & 2 \\ 4 & -2 \end{pmatrix}$ then $B^T = \begin{pmatrix} 0 & 4 \\ 2 & -2 \end{pmatrix}$

The **identity** matrix of size n is an $n \times n$ diagonal matrix, denoted I_n, where every entry is 0, except for the leading diagonal, where all entries are 1. For example:

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Identity matrices have the property that any matrix multiplied by an identity matrix will remain unchanged (as long as the multiplication can take place).

Eigenvalues and Eigenvectors

To find the inverse of a 2×2 matrix, there is a simple formula.

$$\boxed{\text{If } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ then } A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}}$$

Example:

Let $A = \begin{pmatrix} 2 & -1 \\ 3 & 5 \end{pmatrix}$. Find A^{-1}.

Eigenvalues and Eigenvectors

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Example:
Let $A = \begin{pmatrix} 2 & -1 \\ 3 & 5 \end{pmatrix}$. Find A^{-1}.

Using the formula,

$$A^{-1} = \frac{1}{(2)(5) - (3)(-1)} \begin{pmatrix} 5 & 1 \\ -3 & 2 \end{pmatrix} = \frac{1}{13} \begin{pmatrix} 5 & 1 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} \frac{5}{13} & \frac{1}{13} \\ -\frac{3}{13} & \frac{2}{13} \end{pmatrix}$$

Note that the value $ad - bc$ is called the **determinant** of the matrix A (for a 2×2 matrix).

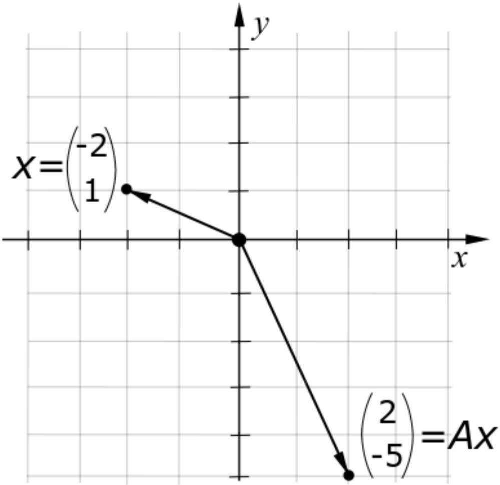
Eigenvalues and Eigenvectors

Earlier we saw that $\begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$.

More generally, a vector multiplied by a matrix will give another vector. In this way we can view matrices as functions on a vector space: they take a vector as input, and give another vector as output. It turns out these represent a special kind of function called a **linear function**.

Lets look at what the input and output of our example look like on a graph:

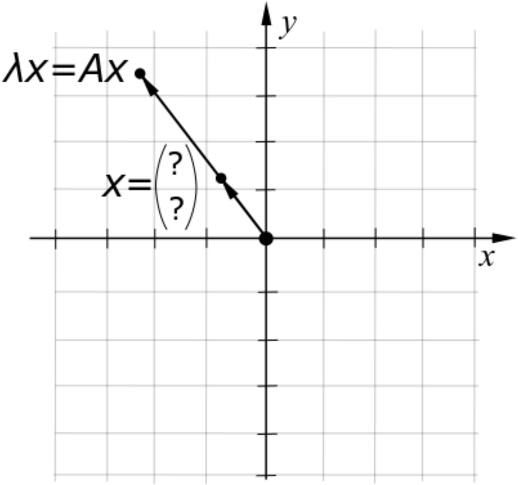
Eigenvalues and Eigenvectors



We input a $\mathbf{x} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, and get vector $A\mathbf{x} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ as an output.

Eigenvalues and Eigenvectors

Question: with our matrix $A \begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix}$, can we find a vector $\mathbf{x}$ such that $A\mathbf{x}$ is a scalar multiple of $\mathbf{x}$?



Eigenvalues and Eigenvectors

This gives the **eigenvector problem**. It can be stated as follows: given a matrix A, find a vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

Such a vector $\mathbf{x}$ is called an **eigenvector**, and the corresponding value λ is its **eigenvalue**.

There is a method to find such eigenvalues and eigenvectors. We will illustrate it for 2×2 matrices (although the theory can be extended for any $n \times n$ matrix - it just becomes a bit technically harder to compute).

Eigenvalues and Eigenvectors

First we need to calculate the eigenvalues. These are found by solving

$$\det(A - \lambda I) = 0.$$

This just means subtracting λ from the diagonal of A. This is best illustrated in our example.

We have

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 4 \\ 2 & -1-\lambda \end{pmatrix}.$$

Then

$$\begin{aligned} det(A - \lambda I) &= (1-\lambda)(-1-\lambda) - (2)(4) \\ &= \lambda^2 - 1 - 8 \\ &= \lambda^2 - 9 \end{aligned}$$

so $\lambda = 3$ and $\lambda = -3$ are our two eigenvalues.

Eigenvalues and Eigenvectors

Now we can find our eigenvectors. There will be one for each eigenvalue. We can find them as follows:

For an eigenvalue λ, we need to find a vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ such that

$$(A - \lambda I)\mathbf{x} = \mathbf{0}.$$

Since we have already found our eigenvalues λ, we have that $(A - \lambda I)$ will just be a matrix with numbers. Lets see how to do this with our example.

Eigenvalues and Eigenvectors

<u>For $\lambda = 3$:</u>
We have

$$A - \lambda I = \begin{pmatrix} 1-3 & 4 \\ 2 & -1-3 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 2 & -4 \end{pmatrix}.$$

We need to find a vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ such that

$$\begin{pmatrix} -2 & 4 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

or

$$\begin{aligned} -2x_1 + 4x_2 &= 0 \\ 2x_1 - 4x_2 &= 0 \end{aligned}$$

This gives two equations and two unknowns, but (if we have done this right) one equation is a scalar multiple of the other, so there is no unique solution.

Eigenvalues and Eigenvectors

It turns out there are infinitely many solutions possible, all scalar multiples of each other. These are all eigenvectors, but we only need one.

The easiest way to proceed is to choose a (non-zero) value for one of the variables (for example $x_1 = 1$), and work out what the other variable will be. The solution we get should work for both equations.

In this case we let $x_1 = 1$, so then

$$\begin{aligned} (-2)(1) + 4x_2 &= 0 \\ x_2 &= \frac{1}{2}. \end{aligned}$$

So $x_1 = 1$ and $x_2 = \frac{1}{2}$ will work. If we have done things correctly, **this should work for the other equation too** (check).

Our first eigenvector corresponding to the eigenvalue $\lambda = 3$ is then $\begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}$.

(note this is not unique - any scalar multiple of $\begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}$ will also be an eigenvector)

Eigenvalues and Eigenvectors

Exercise: Find an eigenvector corresponding to the eigenvalue $\lambda = -3$.

Eigenvalues and Eigenvectors

To summarise, we proceed as follows:

- Find eigenvalues by solving $\det(A - \lambda I) = 0$.
- For each eigenvalue λ, find an eigenvector by solving $(A - \lambda I)\mathbf{x} = \mathbf{0}$. This usually involves setting the value for one variable and finding what the other needs to be.

Exercises

Find eigenvalues and eigenvectors for the following matrices:

- $\begin{pmatrix} 7 & -1 \\ 3 & 3 \end{pmatrix}$

- $\begin{pmatrix} 3 & 1 \\ 6 & 2 \end{pmatrix}$

Text Book References

Chapters 4.0, 4.1